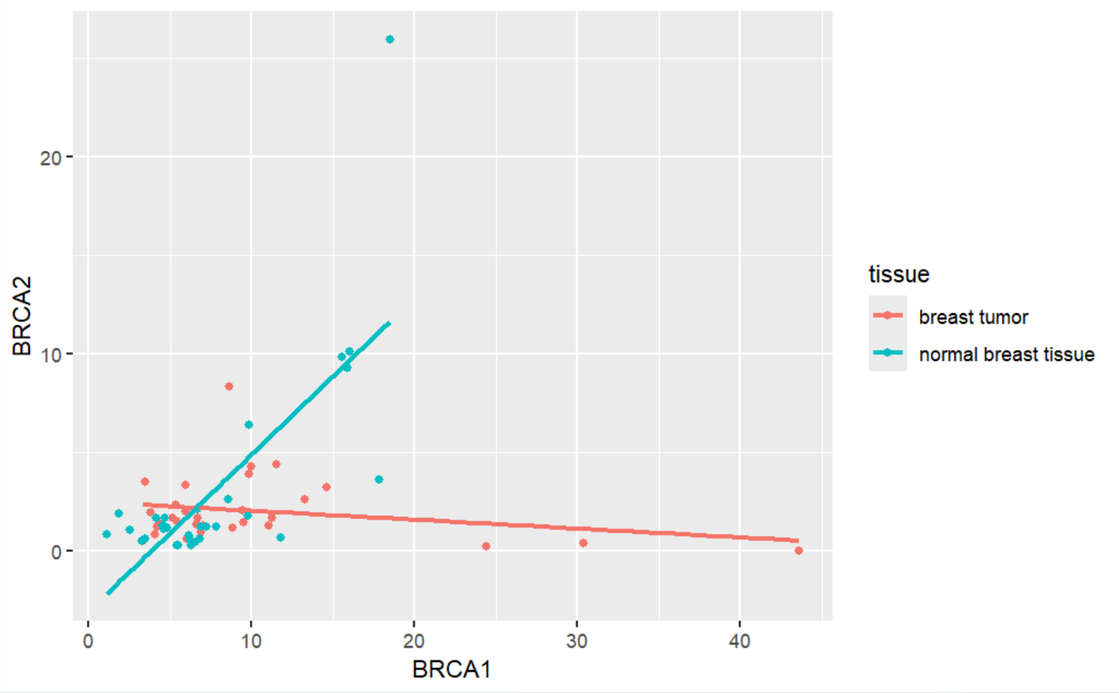
Cancer Research

Data Analysis on Breast Cancer

Date

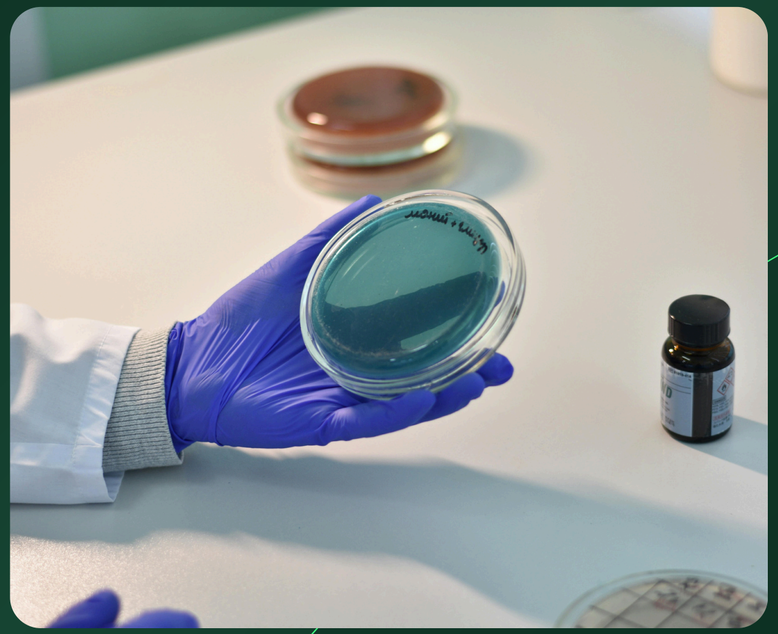
217 MAR, 2026

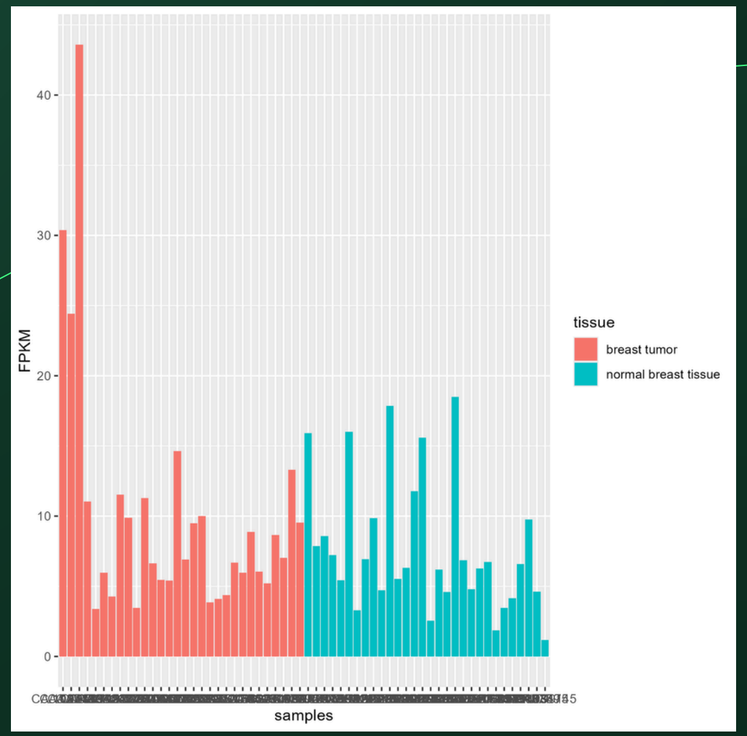


Introduction

Understanding the Basics

- Genes are switched on and off. This is called **expression**.
- How often they are expressed is highly regulated process.
- With Cancer, this system goes haywire.





Aim

Identify which genes are going haywire

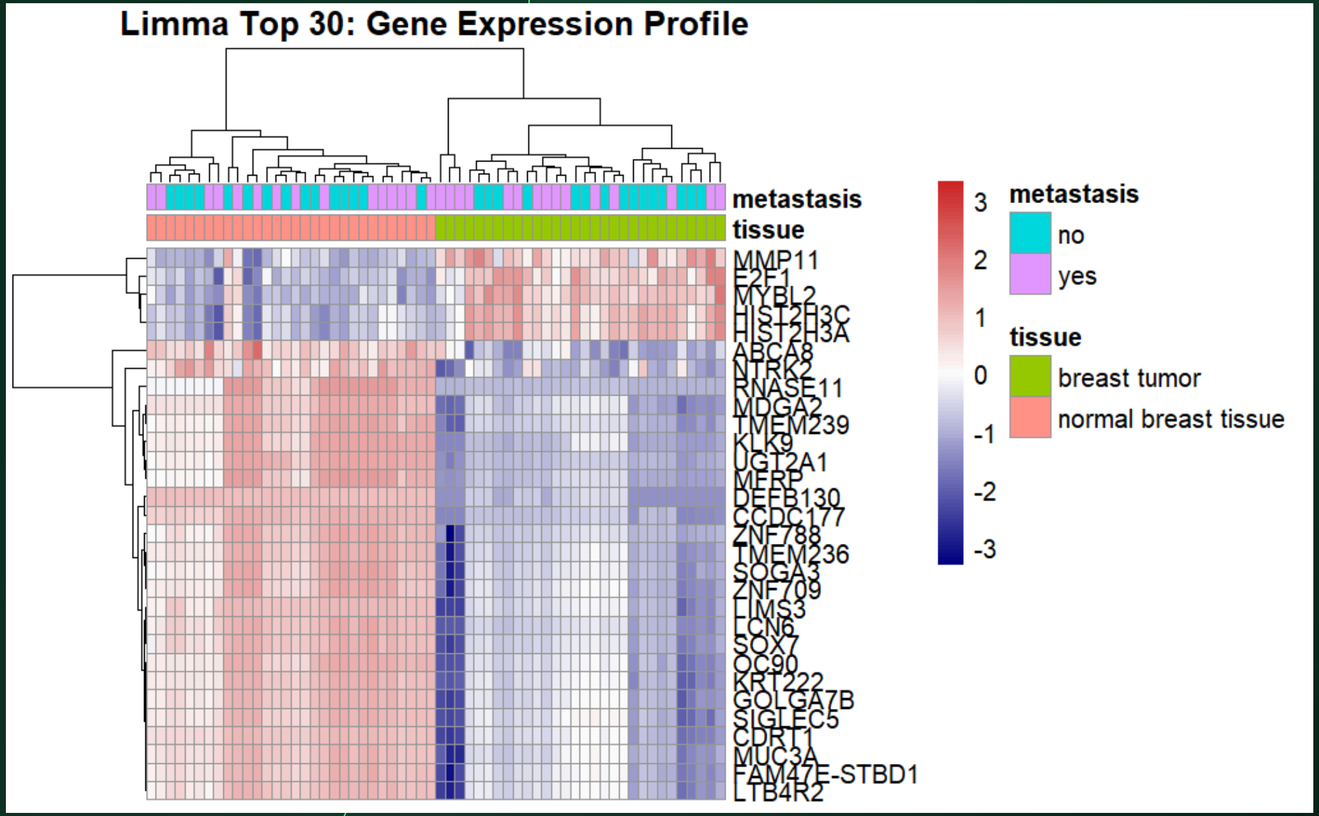
Based on Gene Expression Data,
can we figure out which genes are being expressed differently between cancer and healthy samples?

gene <chr>	tissue <chr>	mean_FPKM <dbl>
CCDC177	breast tumor	1.2085000
CCDC177	normal breast tissue	23.1078333
DEFB130	breast tumor	0.9893333
DEFB130	normal breast tissue	25.6073333
KLK9	breast tumor	1.7303333
KLK9	normal breast tissue	28.5250000
LCN6	breast tumor	5.4883333
LCN6	normal breast tissue	63.8808333
UGT2A1	breast tumor	2.9148333
UGT2A1	normal breast tissue	18.3401667

Heatmap

Identify which genes are going haywire

Differentially expressed genes can be visualised using the Heatmap



Conclusion

The Impact of Bioinformatics

Summary Points

- Now we know that that MMP11, E2F1 and MYBL2 are the top genes that are differentially expressed.
- These genes are over-expressed in the tumor samples.
- If we can identify the function of these genes and develop a drug to regulate these genes, we slow down the proliferation of cancer cells.